Week 4

2) Wireless Sensor Network Energy optimization

Application: Wireless Sensor network.

Problem Statement: A wireless Sensor network contains multiple Sensor nodes with limited battery energy. Develop a CS-based optimization to identify Suitable Communication parameters & routing Configurations that reduce energy Consumption. while maintaining reliable data transmission.

Objective: Minimize network energy Consuptions and extend network lifetime while maintaining an acceptable Packet ~~data~~ delivery rate.

Pseudocode:

Begin

1. Initilize parameters
   Set numbe of nests, discovery probability (Pa), maximum iterations and network parameter.
2. Initilize population.
   Generate random nests representing different routing Configurations and Communication. Parameter.
3. Evaluate Fitness
   Calculate ~~energy~~ energy Consumption and Packet delivery rate for each nest.
   Select Solutions with low energy Sonsumption and acceptable packet delivery rate
4. Generate new Solutions
   generate new solutions using Levy filight
   Evaluate their fitness and Select better Solutions.

5. Abandon worst Nest
Abandon a fraction (Pa) of the worst nests.
Replace them with new random solutions

6. Check and Output
Repeat Steps 3-5 until the maximum iteration are reached.
Display the best routing Configuration, minimum energy Consumption and packet delivery rate.

End

output:

Wireless Sensor Network Energy Optimization

Optimal Transmission power: 0.474
Optimal Communication Distance: 10.0
Minimum Energy Consumption: 5.243
Packet Delivery Rate: 85.0%
Network Lifetime (relative): 19.07

Shetty
30/10/25